

manufacturers to prevent overbuilding, and the expense in legitimately creating a comparable FPGA design from scratch, FPGAs can actually help reduce the occurrence of overbuilding.

An ever-increasing design security concern is the use of counterfeit or untrusted components. Due to the high cost of developing FPGAs, and the controls in place at the FPGA foundries, the chance of getting a functioning FPGA manufactured with an unauthorised mask set is fairly small. Also, since FPGA and cSoC ICs are unique in that a very large part of the IP is programmed into the device by the

this data, or to ensure its integrity and authenticity.

An FPGA or cSoC without solid design security features is not a good candidate for data security applications. In many projects data security is only a minor concern, but in some applications, such as financial payment and military communication systems, data security is paramount. Data security requirements are becoming more prevalent with the explosion of the 'Internet of Things,' i.e., 'things' other than conventional personal computers and servers attached to the Internet. These include industrial control sen-

credentials being stolen from Web servers. However, threats may be closer to home, such as a crooked contract manufacturer or a company insider stealing the company's IP assets.

Finally, there are attacks that get right down to the hardware level, and try to steal the IP right out of a device. This may involve first extracting cryptographic keys from the device that are used to protect the IP, perhaps together with capturing a new configuration file being downloaded to a device in the field. Extracting processor firmware and reverse engineering it is another real threat.

Attacks only get better with time. Security that was considered good enough for many applications a few years ago may now be considered too vulnerable to use in the same applications today.

An example of attack at the device level that is becoming more prevalent and less expensive to perform is 'differential power analysis' (DPA). In it, the power supply consumption of an integrated circuit like a microcontroller or FPGA is monitored while it performs cryptographic operations on data using a secret key. It is assumed that the adversary has access to the encrypted form of the data—or else why encrypt it in the first place? With that data and enough power supply measurements collected during the time the key was being used, the adversary can compute the value of the key even without depackaging the chip. Depending on the rate of information leakage via the power supply and the sophistication of the attack, keys may be extracted in timeframes ranging from days down to milliseconds.

Without countermeasures, all electronic systems performing cryptography are subject to DPA-type attacks. These include most of the microcontrollers and all the FPGAs currently on the market. Reports published over the last several years have demonstrated attacks where the keys used to load encrypted bitstreams into several brands of FPGAs were extracted using DPA. With the keys, it is usually possible then, and in some cases easy,



Design security concepts: Protection against overbuilding (Courtesy: www.actel.com)

OEM, the risk of untrusted IP is largely moved to the configuration step rather than the foundry step.

But, a very real risk is the entry into the supply chain of used parts, or parts that are re-marked from authentic but less expensive versions to more expensive devices that should be designed or screened to faster speed grades, high temperature ranges or higher reliability levels. These parts may even work, part of the time. Certainly, the customer is not receiving the value that was paid for and is potentially risking system failures.

Data security

Data security moves the focus from the OEM's IP to the data processed by the device. This is typically data owned by the OEM's customer, or the customer's customer, rather than the IP owned by the OEM. This is now into the realm of information assurance, i.e., protecting the data stored, processed or communicated by the device during run-time. Often, cryptographic methods are used to protect the confidentiality of

sensors and systems, medical devices, smart electric meters and appliances, and a myriad of wireless devices.

Tampering

Tampering can occur at the device level or system level. This is where an attacker tries to gain some advantage by finding out information to which he shouldn't have access. This may be by eavesdropping on signals that were intended to be confidential, to more active attacks involving changing signals or the state of a system by various means. Often the goal of attackers is to reverse-engineer the system so that they can find other security weaknesses, or to accelerate their own development of similar systems.

Network attacks rightly get a lot of attention since poorly designed protocols or network protections such as firewalls may allow an adversary to steal information from the comfort of an Internet café half-way around the world. We read almost every day about thousands of poorly protected credit card numbers or account login